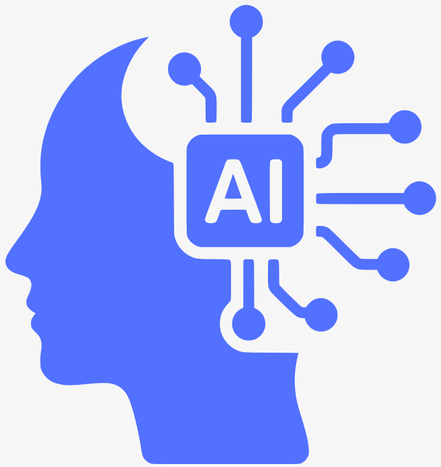
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LANGCHAIN & LANGGRAPH

IMAGINE BUILDING AN AI THAT DOESN'T JUST ANSWER, BUT ACTS: USING TOOLS, REMEMBERING CONTEXT, AND SOLVING COMPLEX TASKS AUTONOMOUSLY.

WHAT IS LANGCHAIN?

- An open-source framework for building applications powered by Large Language Models (LLMs).
- It provides reusable components to connect models, prompts, tools, data sources, and other services.
- It helps developers build AI applications without implementing every integration from scratch.

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WHY DO WE NEED LANGCHAIN?

- Connecting an LLM to external tools and data can require significant integration code.
- LangChain provides reusable interfaces and integrations for common AI development tasks.
- It makes it easier to combine multiple components into an AI application.

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WHAT CAN WE BUILD WITH LANGCHAIN?

- AI chatbots and assistants.
- Retrieval-Augmented Generation (RAG) applications.
- AI applications with function calling and tools.
- Applications that generate structured outputs.

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LANGCHAIN ARCHITECTURE & COMPONENTS

- Models
- Prompts & Prompt Templates
- Chains
- Tools & Tool Calling
- Retrievers
- Structured Output
- Output Parsers

WHAT IS LANGGRAPH?

- LangGraph is a framework for building stateful, multi-step AI workflows and agents.
- It represents application logic as a graph containing nodes and edges.
- It supports cycles, conditional decisions, persistence, and human-in-the-loop interactions.

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WHY DO WE NEED LANGGRAPH?

- Simple linear chains are not always enough for complex AI applications.
- Agents may need to make decisions, repeat actions, and change their execution path.
- LangGraph gives developers more control over how these systems execute.

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WHAT CAN WE BUILD WITH LANGGRAPH?

- Autonomous AI agents.
- Multi-agent systems.
- Stateful conversational assistants.
- Complex AI workflows with conditional routing and human approval.

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LANGGRAPH ARCHITECTURE

- State
- Nodes
- Edges
- Conditional Edges
- Cycles & Loops
- Memory & Persistence
- Human-in-the-Loop

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LANGCHAIN VS LANGGRAPH

HOW THEY WORK TOGETHER

- LangChain can provide the LLM integration, prompt templates, tools, and retrievers.
- LangGraph can coordinate when and how those components are executed.
- Together, they can support intelligent applications that use tools, maintain state, and follow complex execution paths.

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WHEN SHOULD WE USE WHICH?

- Use LangChain: When you need model integrations, prompts, tools, structured outputs, or RAG components.
- Use LangGraph: When you need conditional routing, state management, cycles, persistence, or human approval.
- Use Both: When you want reusable AI components combined with controlled, stateful agent execution.